SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I S. Harsha Vardhan Reddy (192325029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
S. Harsha Vardhan Reddy

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

   a. Show how each attack leads to specific threats and risks
   b. Map how security goals help mitigate them Provide one real-world example for each link

Ans **1. Concept Map: Security Attacks, Threats, Risks and Security Goals (CIA)**

Concept Map

```
                    SECURITY ATTACKS
                           |
        ┌──────────────────┼──────────────────┐
        |                  |                  |
  Passive Attack     Active Attack      Malware Attack
        |                  |                  |
        ▼                  ▼                  ▼
  Information         Data Modification    System Damage
  Disclosure          & Disruption         & Data Theft
        |                  |                  |
        ▼                  ▼                  ▼
      THREATS            THREATS            THREATS
        |                  |                  |
        ▼                  ▼                  ▼
    Unauthorized     Loss of Integrity   Loss of Availability
       Access
        |                  |                  |
        └──────────────┬───┴──────────────────┘
                       ▼
                     RISKS
                       |
        ┌──────────────┼──────────────┐
        ▼              ▼              ▼
  Financial Loss  Reputation Loss  Legal Issues
                       |
                       ▼
              SECURITY GOALS (CIA)
        ┌──────────────┼──────────────┐
        ▼              ▼              ▼
  Confidentiality   Integrity     Availability
  Encryption       Hashing       Backup, Redundancy
  Access Control   Digital Sign. Recovery Systems
```

a. Attack → Threat → Risk

Attack	Threat	Risk
Phishing	Credential Theft	Financial Loss

Attack	Threat	Risk
Data Tampering	Loss of Integrity	Wrong Decisions
DDoS Attack	Service Unavailable	Business Loss
Malware	Data Corruption	System Failure

b. Security Goals Mitigation

Security Goal	Mitigation
Confidentiality	Encryption, Passwords
Integrity	Hash Functions, Digital Signatures
Availability	Backup, Firewalls, Load Balancing

Real-World Examples

Link	Example
Phishing → Credential Theft	Fake banking emails stealing passwords
Malware → Data Loss	WannaCry ransomware
DDoS → Service Outage	GitHub DDoS attack
Encryption → Confidentiality	WhatsApp End-to-End Encryption
Hashing → Integrity	Software checksum verification
Backup → Availability	Google Cloud backups

Analysis

Security attacks create threats that increase organizational risks. The CIA triad provides protection by ensuring information remains confidential, accurate, and accessible. Combining all three goals offers comprehensive security.

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

Ans. **Concept Map: Substitution Techniques**

Concept Map

```
               SUBSTITUTION TECHNIQUES
                        |
   ┌──────────┬─────────────┬─────────────┬─────────────┐
   |          |             |             |
 Caesar   Monoalphabetic Vigenère      Hill Cipher
   |          |             |             |
 Single     Single        Multiple      Matrix Key
 Shift Key  Mapping       Keys          Encryption
   |          |             |             |
 Easy       Moderate      Complex     Very Complex
   |          |             |             |
 Weak       Medium       Strong      Strongest
```

Comparison

Technique	Key Usage	Complexity	Security
Caesar	Single Shift	Very Low	Weak
Monoalphabetic	One Alphabet Mapping	Medium	Moderate
Vigenère	Multiple Keys	High	Strong
Hill Cipher	Matrix Key	Very High	Very Strong

Similarities

- All replace plaintext characters.
- Require encryption and decryption keys.
- Classical symmetric encryption methods.

Differences

Caesar	Monoalphabetic	Vigenère	Hill
Shift letters	Random substitution	Multiple alphabets	Matrix multiplication

Analysis

Most Secure: Hill Cipher

Reasons

- Uses matrix mathematics.
- Encrypts blocks instead of single letters.
- Resistant to frequency analysis.
- Larger key space.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why

Ans. **Concept Map: Classical Encryption Schemes (10 Marks)**

Concept Map

```
          CLASSICAL ENCRYPTION
                  |
        ┌─────────┴──────────┐
        ▼                    ▼
  Substitution         Transposition
        |                    |
 Changes Symbols      Changes Position
        |                    |
 Meaning Changes     Order Changes
        |                    |
        └──────────┬─────────┘
                   ▼
          Ciphertext Produced
                   |
          Improves Security
```

Similarities

- Both hide plaintext.
- Require keys.
- Classical encryption methods.

Differences

Substitution	Transposition
Letters replaced	Letters rearranged
Frequency changes	Frequency remains
Meaning changes	Position changes

Data Structure

Technique	Data Structure
Substitution	Character values change
Transposition	Character order changes

Security Level

Technique	Security
Simple Substitution	Medium
Monoalphabetic	Moderate
Columnar Transposition	Moderate

Vulnerability Analysis
More Vulnerable
Substitution Cipher
Reasons:

- Frequency analysis.
- Letter patterns remain.
- Easily broken with statistical attacks.

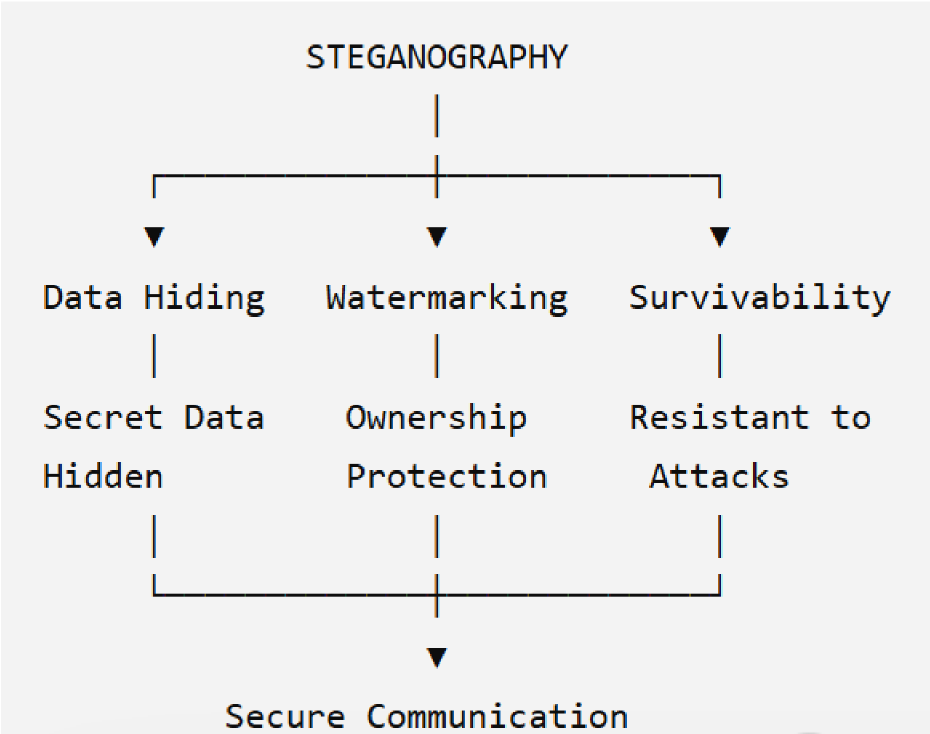
Transposition hides order but preserves frequency, making attacks different but still possible.

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

a. Show relationships between these concepts
b. Analyze how each contributes to secure communication
c. Include one real-world application for each

Ans **Concept Map: Steganography**

Concept Map



Relationships

Concept	Relationship
Data Hiding	Conceals secret information
Watermarking	Protects ownership
Survivability	Ensures hidden data survives attacks

Contribution

Concept	Contribution
Data Hiding	Confidential communication
Watermarking	Copyright protection
Survivability	Reliability after compression or editing

Applications

Concept	Example
Data Hiding	Hidden military communication
Watermarking	Copyright protection of digital photos
Survivability	Secure medical image transmission

Analysis

Steganography provides invisible communication. Watermarking proves ownership, while survivability ensures hidden information remains intact even after modifications.

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

a. Show how each concept is mathematically connected
b. Map their role in encryption/decryption
c. Analyze which concept is most critical for public key cryptography and justify

Ans Concept Map: Number Theory in Cryptography

Concept Map

```
                NUMBER THEORY
                      |
        ┌─────────────┼─────────────┐
        ▼             ▼             ▼
 Extended Euclid  Modular Inverse  Euler Phi
        |             |             |
 Finds GCD      Computes Inverse   Counts Coprimes
        |             |             |
        └─────────────┼─────────────┘
                      ▼
               Euler's Theorem
                      |
                      ▼
          Fermat's Little Theorem
                      |
                      ▼
        RSA Encryption & Decryption
```

Mathematical Connections

Concept	Connection
Extended Euclid	Computes GCD and inverse
Modular Inverse	Needed for private key
Euler Phi	Used in RSA key generation
Euler Theorem	Basis of RSA correctness
Fermat Theorem	Special case of Euler theorem

Role in Encryption

Concept	Role
Extended Euclid	Finds decryption key
Modular Inverse	Generates private key
Euler Phi	Computes RSA parameters

Concept	Role
Euler Theorem	Enables encryption/decryption
Fermat Theorem	Simplifies modular exponentiation

Most Critical Concept
Euler's Phi Function
Justification

- Used to generate RSA public and private keys.
- Determines valid encryption and decryption exponents.
- Essential for Euler's theorem.
- Without $\varphi(n)$, RSA key generation is impossible.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation